

Program-4 : Integrate Git with Docker for Source-Controlled Application Builds

Description :

This program demonstrates integrating Git with Docker to enable source-controlled application builds and deployment using Docker containers.

Step-1 : Create Project Directory

```
cd ~  
mkdir program-4  
cd program-4
```

Step-2 : Initialize Node Project

```
npm init -y
```

Step-3 : Create index.js

```
nano index.js
```

```
const express = require("express");  
const app = express();  
const PORT = 3000;  
  
app.get("/", (req, res) => {  
  res.send("Hello from Git + Docker Integration!");  
});  
  
app.listen(PORT, () => {  
  console.log(`Server running at http://localhost:${PORT}`);  
});
```

Step-4 : Install Dependency

```
npm install express
```

Step-5 : Create Dockerfile

```
nano Dockerfile
```

```
FROM node:18 AS build  
WORKDIR /app  
COPY package*.json ./  
RUN npm install  
COPY . .
```

```
FROM node:18-slim
```

```
WORKDIR /app
COPY --from=build /app /app
EXPOSE 3000
CMD ["node", "index.js"]
```

Step-6 : Create GitHub Repository

Create a new repository on GitHub with name Program_4

Step-7 : Initialize Git Locally

```
git init
git add .
git commit -m "Initial commit"
git branch -M main
```

Step-8 : Connect to GitHub Repository

```
git remote add origin https://github.com/Pavanbsc/Program_4.git
```

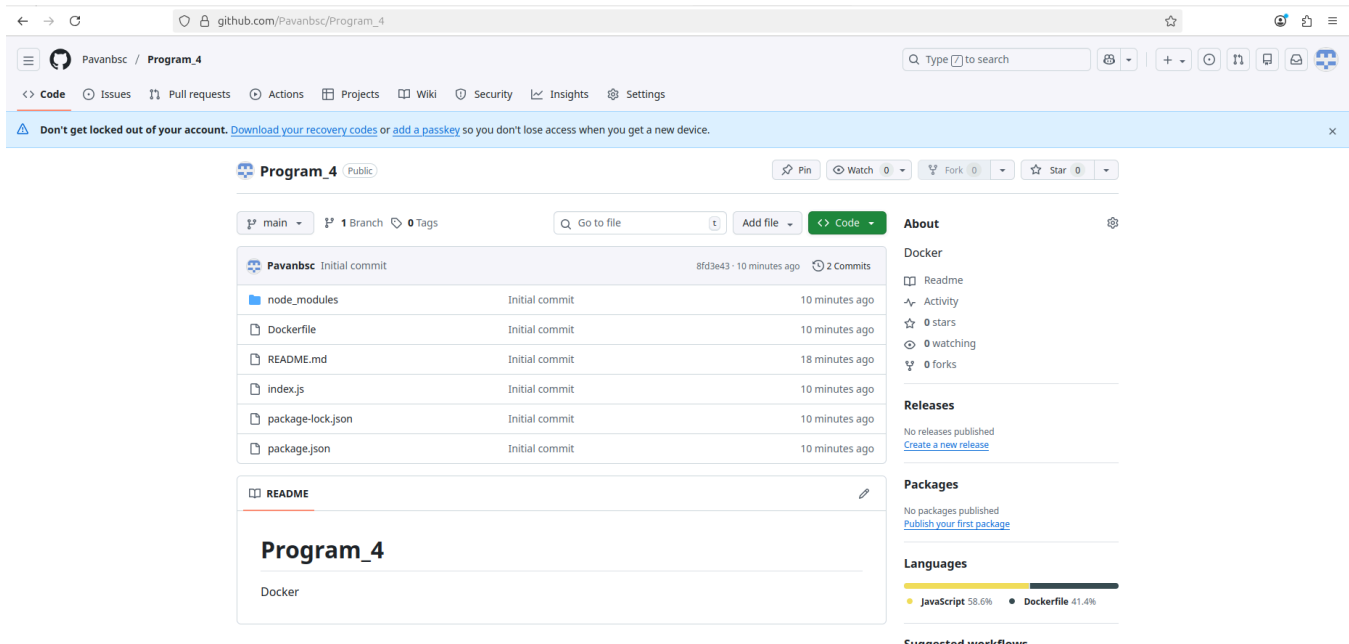
Step-9 : Pull Remote Changes

```
git pull origin main --rebase
```

Step-10 : Push Code to GitHub

```
git push -u origin main
Username : Pavanbsc
Password : GitHub Personal Access Token
```

```
-----
pavan-k@pavank1RV24MC072:~/program-4$ git push -u origin main
Username for 'https://github.com': Pavanbsc
Password for 'https://Pavanbsc@github.com':
Enumerating objects: 679, done.
Counting objects: 100% (679/679), done.
Delta compression using up to 12 threads
Compressing objects: 100% (627/627), done.
Writing objects: 100% (678/678), 670.36 KiB | 9.58 MiB/s, done.
Total 678 (delta 120), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (120/120), done.
To https://github.com/Pavanbsc/Program_4.git
   de373ae..8fd3e43  main -> main
branch 'main' set up to track 'origin/main'.
```



Step-11 : Build Docker Image

`sudo docker build -t program-4 .`

Step-12 : Run Docker Container

`sudo docker run -p 3000:3000 program-4`

```
pavan-k@pavank1RV24MC072:~/program-4$ sudo docker run -p 3000:3000 program-4
Server running at http://localhost:3000
```

Step-13 : Verify Output

`http://localhost:3000`

Hello from Git + Docker Integration!



Step-14 : Stop and Remove Container and Image

`sudo docker ps`

`sudo docker stop <container-id>`

`sudo docker rm <container-id>`

`sudo docker image rm program-4`

Result :

Git is successfully integrated with Docker and the application is built, version-controlled, and executed inside a Docker container.